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```
import java.awt.*;
import java.awt.event.*;

public class Main1 extends Frame implements ActionListener {

    CardLayout cardLayout;
    Panel cards;

    // Customer details
    TextField customerName;
    TextField tableNumber;

    // Food selection
    Choice category;
    Choice foodItem;
    Choice quantity;

    // Bill
    TextArea orderArea;
    TextArea billArea;

    Button addItem;
    Button generateBill;
    Button clear;
    Button next;
    Button previous;

    double subtotal = 0;
    final double GST_RATE = 0.05;

    // Food names and prices
    String[] foodNames = {
        "Pizza",
        "Burger",
        "French Fries",
        "Biryani",
```

```
"Fried Rice",  
"Noodles",  
"Coke",  
"Fresh Juice",  
"Ice Cream"  
};
```

```
double[] foodPrices = {  
    250,  
    120,  
    80,  
    180,  
    150,  
    140,  
    50,  
    80,  
    100  
};
```

```
// Constructor
```

```
public Main1() {  
  
    setTitle("Food Ordering and Billing System");  
    setSize(800, 600);  
    setLayout(new BorderLayout());  
  
    cardLayout = new CardLayout();  
    cards = new Panel(cardLayout);  
  
    createCustomerCard();  
    createFoodCard();  
    createBillCard();  
  
    add(cards, BorderLayout.CENTER);  
  
    setVisible(true);  
  
    addWindowListener(new WindowAdapter() {  
        public void windowClosing(WindowEvent e) {  
            dispose();  
        }  
    });  
}
```

```

    }
    });
}

// -----
// CUSTOMER CARD
// -----

void createCustomerCard() {

    Panel panel = new Panel(new GridBagLayout());

    GridBagConstraints gbc = new GridBagConstraints();

    Label title = new Label(
        "CUSTOMER / ORDER DETAILS",
        Label.CENTER
    );

    title.setFont(
        new Font("Arial", Font.BOLD, 22)
    );

    customerName = new TextField(25);
    tableNumber = new TextField(25);

    next = new Button("Next");

    next.addActionListener(this);

    gbc.insets = new Insets(15, 15, 15, 15);

    // Title
    gbc.gridx = 0;
    gbc.gridy = 0;
    gbc.gridwidth = 2;

    panel.add(title, gbc);

    gbc.gridwidth = 1;

```

```

// Customer Name
gbc.gridx = 0;
gbc.gridy = 1;

panel.add(
    new Label("Customer Name:"),
    gbc
);

gbc.gridx = 1;

panel.add(customerName, gbc);

// Table Number
gbc.gridx = 0;
gbc.gridy = 2;

panel.add(
    new Label("Table Number:"),
    gbc
);

gbc.gridx = 1;

panel.add(tableNumber, gbc);

// Next button
gbc.gridx = 1;
gbc.gridy = 3;

panel.add(next, gbc);

cards.add(panel, "customer");
}

// -----
// FOOD CARD
// -----

```

```
void createFoodCard() {

    Panel panel = new Panel(new BorderLayout());

    Label title = new Label(
        "FOOD SELECTION",
        Label.CENTER
    );

    title.setFont(
        new Font("Arial", Font.BOLD, 22)
    );

    // Input panel
    Panel inputPanel =
        new Panel(new GridLayout(4, 2, 10, 10));

    category = new Choice();

    category.add("Main Course");
    category.add("Rice Items");
    category.add("Beverages");
    category.add("Desserts");

    foodItem = new Choice();

    quantity = new Choice();

    for (int i = 1; i <= 10; i++) {
        quantity.add(String.valueOf(i));
    }

    addItem = new Button("Add Item");
    clear = new Button("Clear");
    previous = new Button("Previous");
    generateBill = new Button("Generate Bill");

    addItem.addActionListener(this);
    clear.addActionListener(this);
    previous.addActionListener(this);
}
```

```
generateBill.addActionListener(this);

// Category change event
category.addItemListener(
    new ItemListener() {
        public void itemStateChanged(ItemEvent e) {
            updateFoodItems();
        }
    }
);

inputPanel.add(
    new Label("Food Category:")
);

inputPanel.add(category);

inputPanel.add(
    new Label("Food Item:")
);

inputPanel.add(foodItem);

inputPanel.add(
    new Label("Quantity:")
);

inputPanel.add(quantity);

inputPanel.add(addItem);
inputPanel.add(clear);

// Order area
orderArea = new TextArea(12, 60);

orderArea.setEditable(false);

// Bottom buttons
Panel bottom = new Panel();
```

```

        bottom.add(previous);
        bottom.add(generateBill);

        panel.add(
            title,
            BorderLayout.NORTH
        );

        panel.add(
            inputPanel,
            BorderLayout.CENTER
        );

        panel.add(
            orderArea,
            BorderLayout.SOUTH
        );

        panel.add(
            bottom,
            BorderLayout.PAGE_END
        );

        cards.add(panel, "food");

        updateFoodItems();
    }

// -----
// BILL CARD
// -----

void createBillCard() {

    Panel panel =
        new Panel(new BorderLayout());

    Label title = new Label(
        "BILL / ORDER SUMMARY",
        Label.CENTER
    );

```

```
);

title.setFont(
    new Font("Arial", Font.BOLD, 22)
);

billArea = new TextArea(20, 70);

billArea.setEditable(false);

Button back = new Button("Previous");

back.addActionListener(this);

panel.add(
    title,
    BorderLayout.NORTH
);

panel.add(
    billArea,
    BorderLayout.CENTER
);

panel.add(
    back,
    BorderLayout.SOUTH
);

cards.add(panel, "bill");
}

// -----
// UPDATE FOOD ITEMS
// -----

void updateFoodItems() {

    foodItem.removeAll();
```



```

String selected =
    category.getSelectedItem();

if (selected.equals("Main Course")) {

    foodItem.add("Pizza");
    foodItem.add("Burger");
    foodItem.add("French Fries");

}

else if (selected.equals("Rice Items")) {

    foodItem.add("Biryani");
    foodItem.add("Fried Rice");
    foodItem.add("Noodles");

}

else if (selected.equals("Beverages")) {

    foodItem.add("Coke");
    foodItem.add("Fresh Juice");

}

else if (selected.equals("Desserts")) {

    foodItem.add("Ice Cream");
}
}

// -----
// GET PRICE
// -----

double getPrice(String item) {

    for (int i = 0;
        i < foodNames.length;

```

```

        i++) {

        if (foodNames[i].equals(item)) {

            return foodPrices[i];
        }
    }

    return 0;
}

// -----
// ADD FOOD ITEM
// -----

void addFoodItem() {

    String item =
        foodItem.getSelectedItem();

    if (item == null ||
        item.equals("")) {

        showMessage(
            "Please select a food item."
        );

        return;
    }

    int qty;

    try {

        qty = Integer.parseInt(
            quantity.getSelectedItem()
        );

    }

```

```
catch (Exception e) {

    showMessage(
        "Please select a valid quantity."
    );

    return;
}

if (qty <= 0) {

    showMessage(
        "Quantity must be greater than zero."
    );

    return;
}

double price = getPrice(item);

double cost = price * qty;

subtotal = subtotal + cost;

orderArea.append(
    item +
    " Qty: " +
    qty +
    " Price: Rs." +
    price +
    " Cost: Rs." +
    cost +
    "\n"
);
}

// -----
// GENERATE FINAL BILL
// -----
```

```

void generateFinalBill() {

    if (customerName.getText()
        .trim().equals("")) {

        showMessage(
            "Please enter customer name."
        );

        cardLayout.show(
            cards,
            "customer"
        );

        return;
    }

    if (tableNumber.getText()
        .trim().equals("")) {

        showMessage(
            "Please enter table number."
        );

        cardLayout.show(
            cards,
            "customer"
        );

        return;
    }

    if (subtotal == 0) {

        showMessage(
            "Please add at least one food item."
        );

        cardLayout.show(
            cards,

```

```

        "food"
    );

    return;
}

double gst =
    subtotal * GST_RATE;

double finalAmount =
    subtotal + gst;

billArea.setText("");

billArea.append(
    "=====\\n"
);

billArea.append(
    "    RESTAURANT BILL\\n"
);

billArea.append(
    "=====\\n\\n"
);

billArea.append(
    "Customer Name : " +
    customerName.getText() +
    "\\n"
);

billArea.append(
    "Table Number : " +
    tableNumber.getText() +
    "\\n\\n"
);

billArea.append(
    "-----\\n"

```

);

```
billArea.append(  
    "ORDER DETAILS\n"  
);
```

```
billArea.append(  
    "-----\n"  
);
```

```
billArea.append(  
    orderArea.getText()  
);
```

```
billArea.append(  
    "-----\n"  
);
```

```
billArea.append(  
    String.format(  
        "Subtotal    : Rs. %.2f\n",  
        subtotal  
    )  
);
```

```
billArea.append(  
    String.format(  
        "GST (5%%)    : Rs. %.2f\n",  
        gst  
    )  
);
```

```
billArea.append(  
    "-----\n"  
);
```

```
billArea.append(  
    String.format(  
        "FINAL BILL   : Rs. %.2f\n",  
        finalAmount  
    )  
);
```

```

    )
);

billArea.append(
    "-----\n"
);

billArea.append(
    "\nThank you! Visit Again.\n"
);

cardLayout.show(
    cards,
    "bill"
);
}

// -----
// CLEAR ORDER
// -----

void clearOrder() {

    subtotal = 0;

    orderArea.setText("");

    foodItem.select(0);

    quantity.select(0);
}

// -----
// BUTTON EVENTS
// -----

public void actionPerformed(
   (ActionEvent e
) {

```

```

Object source =
    e.getSource();

if (source == next) {

    cardLayout.show(
        cards,
        "food"
    );
}

else if (source == previous) {

    cardLayout.show(
        cards,
        "customer"
    );
}

else if (source == addItem) {

    addFoodItem();
}

else if (source == generateBill) {

    generateFinalBill();
}

else if (source == clear) {

    clearOrder();
}
}

// -----
// MESSAGE DIALOG
// -----

void showMessage(

```



```
String message
) {

    Dialog dialog =
        new Dialog(
            this,
            "Message",
            true
        );

    dialog.setLayout(
        new FlowLayout()
    );

    dialog.add(
        new Label(message)
    );

    Button ok =
        new Button("OK");

    ok.addActionListener(
        new ActionListener() {

            public void actionPerformed(
                ActionEvent e
            ) {

                dialog.dispose();
            }
        }
    );

    dialog.add(ok);

    dialog.setSize(
        350,
        120
    );
```

```
        dialog.setLocationRelativeTo(  
            this  
        );  
  
        dialog.setVisible(true);  
    }  
  
    // -----  
    // MAIN METHOD  
    // -----  
  
    public static void main(  
        String[] args  
    ) {  
  
        new Main1();  
    }  
}
```

Food Ordering and Billing System

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BILL / ORDER SUMMARY

=====

RESTAURANT BILL

=====

Customer Name : Ganeshram

Table Number : 145

=====

ORDER DETAILS

=====

Fried Rice Qty: 2 Price: Rs.150.0 Cost: Rs.300.0

Fried Rice Qty: 2 Price: Rs.150.0 Cost: Rs.300.0

Fried Rice Qty: 2 Price: Rs.150.0 Cost: Rs.300.0

Subtotal :Rs. 900.00

GST (5%) :Rs. 45.00

=====

FINAL BILL :Rs. 945.00

=====

Thank you! Visit Again.

Previous

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12:57 PM
8/23/2026

Food Ordering and Billing System

BILL / ORDER SUMMARY

=====

RESTAURANT BILL

=====

Customer Name : Ganeshram

Table Number : 145

=====

ORDER DETAILS

=====

Fried Rice Qty: 2 Price: Rs.150.0 Cost: Rs.300.0

Fried Rice Qty: 2 Price: Rs.150.0 Cost: Rs.300.0

Fried Rice Qty: 2 Price: Rs.150.0 Cost: Rs.300.0

Subtotal :Rs. 900.00

GST (5%) :Rs. 45.00

=====

FINAL BILL :Rs. 945.00

=====

Thank you! Visit Again.

FOOD SELECTION

Food Category:

Beverages

Food Item:

Fresh Juice

Quantity:

7

Add Item

Clear